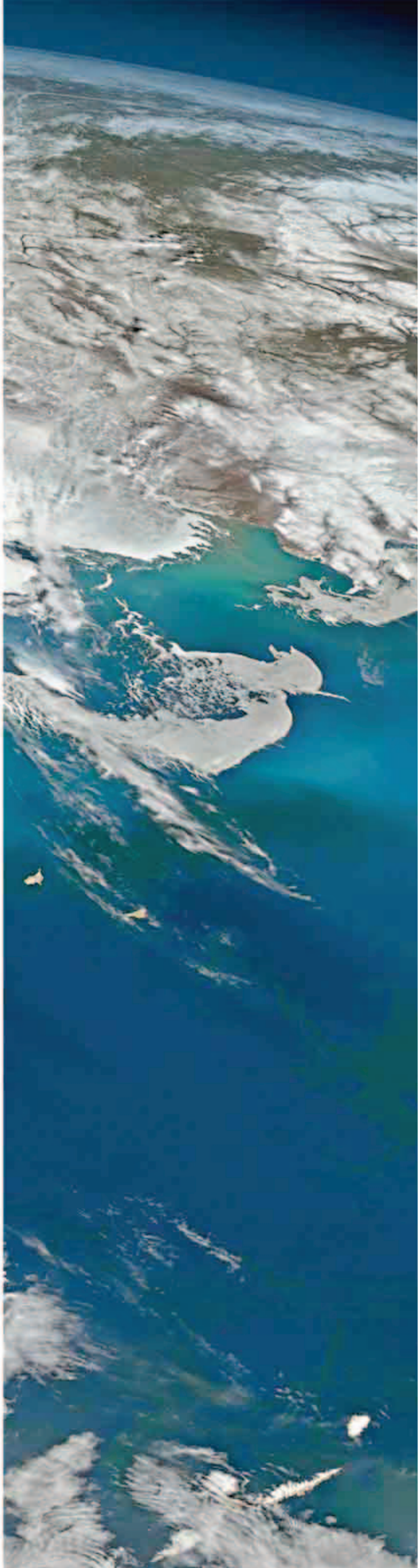
Grinding plates

The process that forms deep ocean trenches also creates long chains of volcanic islands. It pushes up mountain ranges on the fringes of nearby continents, and causes devastating earthquakes and tsunamis. Most of these volcanic subduction zones lie around the edge of the Pacific Ocean, in a region sometimes called the Pacific Ring of Fire.

ISLAND ARCS

Many of the volcanoes erupting from the plate margin rise above the waves. They form a chain of volcanic islands along the line of the plate boundary, known as an island arc. Over time, the islands multiply and join together. A chain of islands such as the Aleutians in the north Pacific will eventually turn into bigger, longer islands such as Java on the Sunda Arc in Indonesia.



▸ ALEUTIAN ARC
The Aleutian Arc, seen here from space, consists of about 70 volcanic islands.

▴ ERUPTION
Krakatau volcano near Java lies above one of the world's most active subduction zones.

MELTDOWN

In a subduction zone, a plate of ocean crust plunges beneath the edge of another plate and into the hot mantle. As it does so, some of the rock melts in the intense heat. This is because ocean water carried down with the rock makes it melt more easily. The molten magma bubbles up through the margin of the upper plate, erupting as volcanoes.

Mount Fitz Roy, southern Andes

FOLDED CONTINENTS

Where ocean floor is being dragged beneath the edge of a continent, the friction buckles the continental fringe into mountain ranges. The Andes on the western side of South America were formed like this, and are dotted with volcanoes erupting from the zone of melting rock deep below the continent.